The University of Faisalabad

COMPUTER VISION

PROJECT REPORT

Registration NUMBER:
2023-BSAI-037
2023-BS-AI-017
2023-BS-AI-015
2023-BS-AI-061

Submitted to:
Ms. Esha Nawaz

Submitted by:
QANDEEL ASIM
HUMNA IMRAN
EMAN TAHIR
AYESHA IMRAN

Department: Artificial intelligence

Skin Disease Classification Using Deep Learning

Abstract
Skin diseases are among the most common health problems worldwide, and early diagnosis is essential for effective treatment. Traditional diagnosis methods require expert dermatologists and manual examination, which can be time-consuming and expensive. This project presents a deep learning-based skin disease classification system using Convolutional Neural Networks (CNNs). The model is trained on skin disease image datasets and can automatically classify different skin conditions from uploaded images. Image preprocessing, data augmentation, and CNN architecture were implemented to improve classification accuracy and model performance. The experimental results show that the developed system can effectively identify multiple skin disease categories with satisfactory accuracy. This project demonstrates the importance of artificial intelligence and computer vision in healthcare applications and medical image analysis.

Introduction
Artificial Intelligence (AI) and Deep Learning have transformed modern healthcare systems by enabling automated disease diagnosis and medical image analysis. Skin diseases such as acne, eczema, psoriasis, melanoma, and fungal infections affect millions of people around the world. Early diagnosis is important to prevent severe complications and improve treatment quality. However, access to dermatologists is limited in many areas, especially in rural regions.

This project aims to develop a skin disease classification system using deep learning techniques. The system uses image processing and Convolutional Neural Networks (CNNs) to analyze skin images and predict disease categories automatically. The project highlights how AI-based solutions can support medical professionals by improving diagnosis speed and accuracy.

Background Study
Deep learning is one of the most successful approaches for image classification tasks. Convolutional Neural Networks (CNNs) are specially designed to process image data and automatically learn important visual features from images. Traditional machine learning techniques required manual feature extraction methods such as edge detection, texture analysis, and color histogram extraction. However, CNNs can automatically learn hierarchical image features directly from datasets.

Many researchers have successfully used CNN architectures such as AlexNet, VGG16, ResNet, MobileNetV2, and EfficientNet for medical image classification tasks. Deep learning systems are widely used in healthcare for disease detection, tumor classification, medical imaging, and diagnostic support systems.

This project applies CNN-based techniques for skin disease classification using medical image datasets and demonstrates the practical implementation of computer vision in healthcare systems.

Methodology

The methodology of this project consists of several important stages including dataset collection, preprocessing, model development, training, testing, and evaluation.

1. Dataset Collection:
A dataset containing multiple categories of skin disease images was collected and organized into folders according to disease classes.

2. Data Preprocessing:
Image preprocessing techniques such as resizing, normalization, and noise reduction were applied before training the model.

3. Data Augmentation:
Data augmentation techniques including image rotation, zooming, flipping, and rescaling were used to increase dataset diversity and reduce overfitting.

4. CNN Model Development:
A Convolutional Neural Network (CNN) architecture was implemented using:
- Convolutional layers
- Max pooling layers
- Dropout layers
- Flatten layer
- Dense fully connected layers
- Softmax output layer

5. Model Training:
The model was trained using the Adam optimizer and categorical cross-entropy loss function. The dataset was divided into training, validation, and testing sets.

6. Prediction System:
A prediction module was developed where users can upload a skin image and receive the predicted disease category automatically.

Results

The developed CNN model successfully classified multiple skin diseases with satisfactory performance. During training, both training accuracy and validation accuracy improved consistently. The model demonstrated the ability to identify visual patterns in medical images effectively.

The project achieved the following outcomes:
- Successful implementation of a CNN-based classification system

- Accurate prediction of multiple skin disease categories
- Reduced overfitting using dropout and data augmentation techniques
- Functional image upload and prediction system

The experimental results confirmed that deep learning models can effectively assist healthcare systems by providing automated diagnostic support for skin disease detection.

Conclusion

This project presented a deep learning-based skin disease classification system using Convolutional Neural Networks. The developed model successfully classified skin diseases from medical images and demonstrated the effectiveness of AI in healthcare applications.

The CNN architecture automatically extracted important image features and improved classification performance. Data preprocessing and augmentation techniques further enhanced the model's accuracy and stability.

Future improvements may include:
- Using larger medical image datasets
- Implementing transfer learning models such as MobileNetV2 or ResNet
- Deploying the project as a web or mobile application
- Adding real-time camera-based disease detection

This project demonstrates the growing importance of artificial intelligence and computer vision in medical diagnosis systems.

References

1. Esteva, A., Kuprel, B., Novoa, R. A., et al. (2017). Dermatologist-Level Classification of Skin Cancer with Deep Neural Networks. Nature, 542(7639), 115–118.

2. LeCun, Y., Bengio, Y., & Hinton, G. (2015). Deep Learning. Nature, 521(7553), 436–444.

3. Goodfellow, I., Bengio, Y., & Courville, A. (2016). Deep Learning. MIT Press.

4. Chollet, F. (2017). Deep Learning with Python. Manning Publications.

5. Krizhevsky, A., Sutskever, I., & Hinton, G. (2012). ImageNet Classification with Deep Convolutional Neural Networks. Advances in Neural Information Processing Systems.

6. TensorFlow Documentation. https://www.tensorflow.org

7. Keras Documentation. https://keras.io